MAXIMILIAN VINCENTIUS

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maximilianvincentius.com | pointxtech.com | phyxhub.com | swastisvarna.org

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EDUCATION

Sekolah Ciputra, IB Diploma **Expected 2027**

- Higher Level (HL): Computer Science, Mathematics AA

UCLA Game Lab Summer Institute **July 2026, Selected**

- Intensive program focusing on Game Design and interactive systems, earning 4.0 UCLA.

TECHNICAL EXPERTISE

Programming Languages: Python, C# (Unity), C++, JavaScript, HTML, CSS

Frameworks & Tools: Unity, Git, Autodesk Maya, Adobe Substance Painter, Node.js, NoSQL, SQL, Supabase, Tailwind

Key Competencies: Full-Stack Web Development, 3D Modeling & Texturing, Game Physics Simulation, Interactive Systems Architecture

ENGINEERING & SYSTEMS

Founder – pointxtech.com (Independent Software Studio) **June 2025 – Present**

- Led development and deployment of multi-platform systems (web, game, simulation) through a unified pipeline, achieving seamless integration and efficient delivery.
- Phyxhub.com (Physics Education Initiative, Aug 2025 – Present): Built a full-stack IB Physics Learning Hub (Vite, React, SQL) featuring a mastery-gated content-locking system and a Figma-designed UI, alongside a library of simplified physics guides and visual aids.
- Fractals of Entropy: Engineered a 3D action-strategy game engine with procedural generation, AI pathfinding, and based combat systems, shipping builds across Windows, Android (Google Play), and mobile.
- Metro Circuit: Built a Unity-based simulation with NPC navigation, traffic systems, and real-time world interaction, shipping builds across Windows, Android (Google Play), and mobile

Lead System Architect – Swastisvarna.org (Non-profit Infrastructure) **January 2024 – Present**

- Rebuilt legacy web infrastructure into a Node.js + Supabase full-stack system deployed on a VPS, including a custom CMS that empowered non-technical staff and cut manual administrative workload by approximately 5 hours per week.

Founder – 3D Game Creator Club **September 2025 – November 2025**

- Rebuilt legacy web infrastructure into a Node.js + Supabase full-stack system deployed on a VPS, including a custom CMS that empowered non-technical staff and cut manual administrative workload by approximately 5 hours per week.

RESEARCH & COMPUTATIONAL INVESTIGATIONS

Complex Fourier Analysis of Instrument Timbre (Mathematics AA HL) **July 2026 – August 2026**

- Decomposed recorded piano and flute/saxophone tones into complex Fourier coefficients(C_n) to test how well the coefficients distinguish instruments at equal pitch across dynamics. Applied the Nyquist sampling theorem, reconstructed recordings from the extracted coefficients, and scored accuracy via Mean Squared Error against the originals, demonstrating applied signal-processing and complex-analysis skills.

Eddy-Current Damping of a Pendulum (Physics HL) **June 2026 – July 2026**

- Built a Waltenhofen-style pendulum apparatus to test how magnetic field strength (B) affects the damping coefficient (γ) of an eddy-current brake. Measured B with a Gaussmeter across five settings, extracted γ from ln(A)-vs-swing-number graphs via video analysis, and confirmed the predicted $\gamma \propto B^2$ relationship from Faraday's and Lenz's laws, providing empirical insight into electromagnetic braking.

Golden-Ratio Wave Cascades in Real-Time Ocean **May 2026 – August 2026**

- Tested whether golden-ratio frequency spacing across Gerstner-wave cascades reduces large-scale periodicity in a real-time Unity HLSL compute-shader ocean, versus a single-cascade baseline and integer-ratio cascades of equal wave count. Implemented the cascades as a GPU compute shader and quantified visual realism via the Structural Similarity Index (SSIM), contributing to optimized, artifact-free real-time water rendering

AWARDS & HONORS

Technology: Gold Medal (6th), Intl. STEM Coding Olympiad (2023); Gold Medal, NCSS Grok Python (2022, 2023).
Mathematics: Silver Medal, Intl. Online Math Challenge (2024); Gold Medal, ASMO Mathematics (2022).
Science: Silver Medal, ASMO Science (2022); Honorable Mention, Intl. STEM Math Olympiad (2023).
Writing/Leadership: Double Gold Medalist (Writing), World Scholar's Cup; TOC Qualifier (2024).